Analyze Incidents & Flights

CS3200 Final Project

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Summer 2, 2025

Analysis by Month

The following will be the table contain the total number of incident by type and month:

Table 1: Total Number of Incidents by Type and Month

Month	crew	faa	mechanical	medical	security	turbulence	weather	wildlife	TSA	Total
January	116	137	124	131	118	102	123	0	0	851
February	202	202	249	218	225	248	225	0	0	1569
March	210	244	223	219	234	221	227	0	0	1578
April	221	245	221	236	251	221	219	0	0	1614
May	239	229	218	229	259	244	271	1	0	1690
June	236	232	244	224	248	212	206	0	0	1602
July	217	253	251	251	232	244	243	0	0	1691
August	241	242	247	214	255	218	242	0	0	1659
September	244	215	241	249	235	261	226	0	0	1671
October	268	233	211	216	231	238	232	0	0	1629
November	230	242	234	252	245	245	221	0	1	1670
December	118	120	113	103	122	113	129	0	0	818

The monthly incident table shows a strong seasonal pattern in bird strike activity. Winter months seems to have a lower activity with December having the least incidents (818 incidents) and July show most acivity with 1691 incidents. In Summer month (May - September), the average 1663 incidents per month while the Winter month (December - February) average only 1079 incidents per month. This is around 54 % increased during the warmer months. The reason to this might due to the bird migration season which makes summer months a more critical periods for preventing bird strike incident.

Analysis by Airline

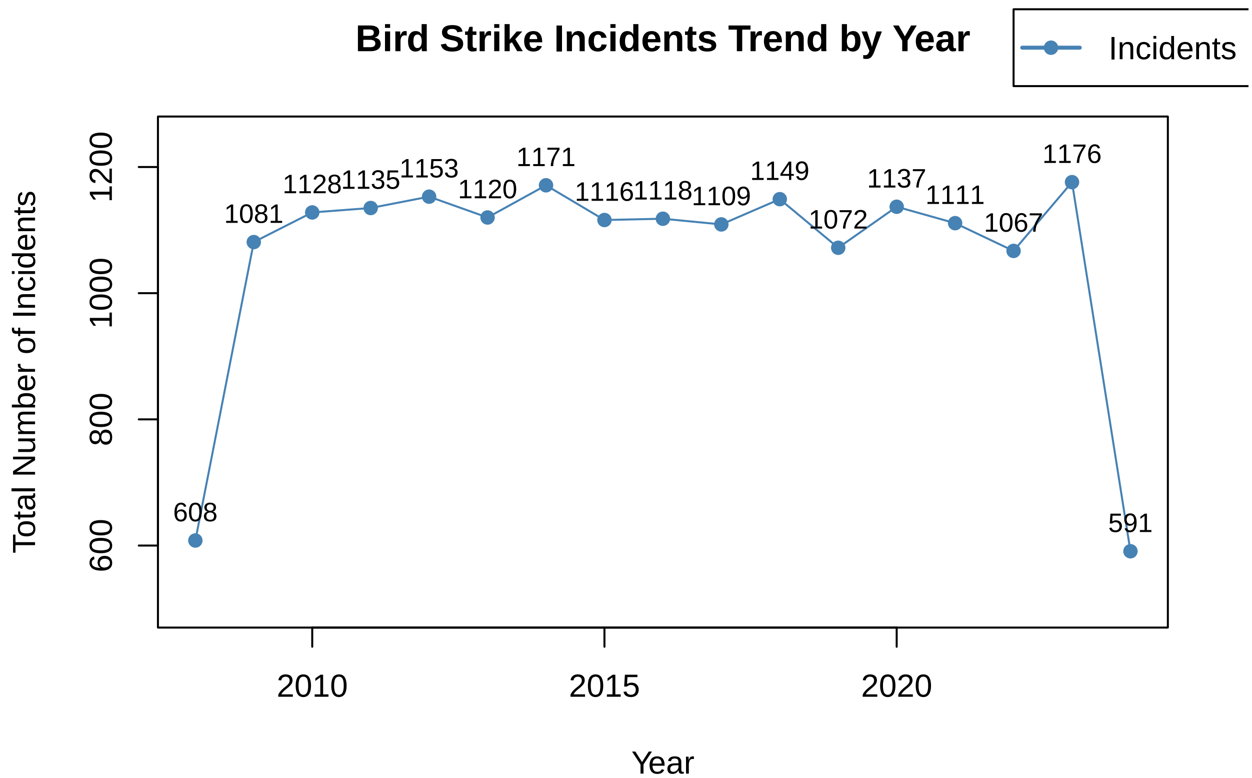
Table 2: Incident Summary and Average Delay by Airline

Airline	Total Incidents	Average Delay (minutes)
AM	509	72.56
AZ	499	72.75
EK	491	72.52
JL	481	72.43

Airline	Total Incidents	Average Delay (minutes)
G3	481	76.49
TG	478	72.34
MH	478	72.44
DL	474	72.85
WN	472	74.58
LA	467	74.59
SQ	463	69.51
UA	462	74.02
KE	458	72.25
AC	456	73.70
LH	455	73.60
DY	454	76.11
AY	453	73.00
EY	450	76.06
IB	450	74.78
CX	449	74.06
CZ	449	75.23
AD	444	71.41
QR	442	72.23
U2	441	73.17
AA	440	72.38
AV	440	72.99
FR	440	75.84
NH	438	73.17
SU	435	74.17
BA	435	72.92
NZ	432	71.88
TK	430	75.40
OZ	429	72.04
MU	428	74.11
AI	428	75.98
KL	428	72.65
SV	427	75.88
AF	424	77.14
QF	423	70.64
VN	409	73.33

The airline analysis shows the performance between airline, with total incidents from 409 (VN) to 509 (AM) with a difference of 100 incidents. This show us that the incident is mostly correlates with operation volume. We observe a average delay vary by 7.6 minutes between the best (SQ at 69.51 minutes) and worst (AF at 77.14 minutes) airline. The small variance shows overall industry protocols for incident response. The differences are likely due to other operational reason than the safety protocol issue.

Trend by Year



The visualization analayze 17 year from 2008 to 2024 showing the trend for airline safety evlotion. It start with 608 in 2008, but in the next year the incidents raise dramatically to 1081 incidents in 2009 and then maintain this level between 1067 - 1171 incidents annually through 2022 with average of 1119.1 incidents per year. In 2023 there is a peak occur with 1176 incidents. And at the very end of the trend, there is a sharp decline to 591 incidents in 2024 which create a situation that the beginning and ending years are having a significantly lower incidents acitivities compare to the middle of the trend. This situation shows that there there are some factor that might cause the significant changes during early and end of the periods. Some posibility might be in the early stage there is a new prevention system is being added to the airline operation and it is taking sometimes for them to adapt to the system in the middle which cause the higher number of incidents and in the end new updates is made into the system and makes it more effective which reduce the incidents activities.